Referee report on "Uniform two-parameter asymptotics for Lanczos coefficients of centered uniform convolutions"

with independent proofs, a partial Lean 4 formalisation and numerical checks

Independent review prepared for Roberto Alfano
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Scope and status. This is a separate research document. It reviews the five-page note of 20 October 2026 (henceforth *the note*) under the instructions of the author. The manuscripts V032 and V035 were not modified. Mathematical validity, bibliographic novelty and formal-machine certification are reported separately, in Sections 1–3, Section 5 and Section 6 respectively. Statements of Levin and Lubinsky, Floater and Milovanović were checked against the original documents supplied by the author (page images were inspected wherever the extracted text was garbled). Numerical computations are consistency checks only; they are not used as proof of anything.

1 Referee's verdict.

Verdict: REPAIRABLE. I believe the theorem is true, and the architecture of the proof is sound: the scaling, the reduction to the Mhaskar–Rakhmanov–Saff equations and the use of Levin–Lubinsky's Theorem 13.4 at the degrees $m = N-1, N$ are correct, and every factor of M, N, 2 and π checks out (Section 3.1). No counterexample was found; on the contrary, high-precision computations up to $M = 400$, $N = 1600$ show $\beta_N^{[M]}/M^2 - F(N/M) = c(N/M)/M + O(M^{-2})$ with a stable function c (Section 4). The note is not VALID as submitted because several steps are asserted rather than proved. The most substantial one is the remainder estimate in the differentiated saddlepoint expansion of Lemma 1, which is the analytic heart of the argument. There are also three smaller defects. Lemma 2 is false as literally stated for $M = 1$. The choice of r^* in Lemma 3 does not account for $a_{N,M} \neq a_t$. Definition 1.1 of Levin–Lubinsky requires a whole sequence indexed by every $n \geq 1$, which the diagonal argument must supply explicitly. None of these gaps is fatal. Section 3 supplies a complete proof of each missing step; with these repairs the theorem is, in my assessment, proved on paper, *conditionally on Theorem 13.4 of Levin–Lubinsky as published*. It is *not* certified by machine in full: the Lean 4 development covers the logical skeleton and every elementary step, but not the Fourier–Laplace analysis of Lemma 1 nor the potential theory behind Levin–Lubinsky's theorem (Section 6).

2 Line-by-line mathematical audit.

Each finding gives the place in the note, the issue, the source checked and a severity: *fatal* (invalidates the result), *substantial* (a step of the proof is missing), *minor* (a correction of statement or wording). No fatal problem was found.

A1. Page 1, definitions (minor, no change needed). The density of $Y_M = X_M/M$ is $g_M(x) = Mf_M(Mx)$. The quantity $\Lambda(s) = \log\{\sinh(s/2)/(s/2)\}$ is the cumulant generating function of Uniform$[-\frac{1}{2}, \frac{1}{2}]$, since $\int_{-1/2}^{1/2} e^{st}\,dt = 2\sinh(s/2)/s$. For the monic recurrence of the note, $P_{N+1} = xP_N - \beta_N P_{N-1}$ (no diagonal term, by symmetry), the orthonormal recurrence is $xp_N = b_{N+1}p_{N+1} + b_N p_{N-1}$ with $b_N = \beta_N^{1/2}$, which is the Nth off-diagonal entry of the Jacobi matrix. This is Milovanović's indexing [7, (2)–(3)], so the $\beta_k^{[m]}$ there and the $\beta_N^{[M]}$ of the note coincide (his weight is f_M translated to $[0, M]$, and translation does not change β). In the theorem, $N \geq L$ and $N/M \in K$ force $M \geq L/\max K$, so both parameters tend to infinity.

A2. Equation (3) of the note, Legendre duality (correct). The identities $I' = s$ and $I'' = 1/v$ hold; they are formalised in Lean (`legendre_deriv`, `legendre_second`).

A3. Lemma 1, Fourier representation (6) (minor). The formula is correct for $M \geq 2$. The cleanest justification needs no contour shift: the tilted law with density $e^{sy}f_M(y)/A(s)^M$ has characteristic function $\{A(s+iu)/A(s)\}^M$, which is integrable for $M \geq 2$ because $|A(s+iu)| \leq 2\cosh(s/2)/|u|$, and Fourier inversion applies. The note gives no justification; Section 3.2 supplies it.

A4. Lemma 1, analytic logarithm and (7) (minor). The zeros of $A(z) = 2\sinh(z/2)/z$ are $z = 2\pi ik$, $k \neq 0$, so A does not vanish on a strip $\{\Re z \in S',\ |\Im z| \leq \varepsilon\}$ for every compact S' and $\varepsilon < 2\pi$; for small ε the values of A there have positive real part and the principal logarithm is analytic. The note asserts this without proof; it is true.

A5. Lemma 1, derivative cancellations (correct). With $s'v = 1$, the coefficient of iu in $\partial_x\Psi$ is $s'v - 1 = 0$, and the coefficient of iu in $\partial_x^2\Psi$ is $s''v + s'^2\Lambda'''(s) = 0$. Both identities are proved in Lean (`tilt_first`, `tilt_second`, `tilt_cancellations`). Hence $\partial_x\Psi = O_J(u^2)$ and $\partial_x^2\Psi = O_J(u^2)$, as claimed.

A6. Lemma 1, outer region (substantial, repairable). The bounds $\rho_{J,\varepsilon,R} < 1$ and $|A(s+iu)/A(s)| \leq C_J/(1+|u|)$ are correct. The claim that ∂_x^j of the integrand is "bounded by a polynomial in M and $1+|u|$ times an $(M-j)$th power" is correct but unproved. The required estimates are $|\partial_x G| \leq C$ and $|\partial_x^2 G| \leq C(1+|u|)$ for $G = e^{-iux}A(s(x)+iu)/A(s(x))$; they are proved in Section 3.2. The justification of differentiating under the integral sign twice is not mentioned; it follows from the same majorants for $M \geq 7$.

A7. Lemma 1, inner region (substantial, repairable). The expansion of $e^{M\Psi}$ is correct, including the coefficients. The bound $C_J M^{-3/2}(1+|z|^{12})e^{c_J z^2/M^{1/3}}$ for $\mathcal{E}_M$ and its two x-derivatives is stated without proof, and the exponential factor is unexplained (it is $O(1)$ on $|z| \leq M^{1/12}$ and unnecessary). This is the decisive estimate of the lemma, so its absence is a real gap. Section 3.2 proves the bound $C(1+|z|)^{18}M^{-3/2}$. The vanishing of the $M^{-1/2}$ term, also after x-differentiation, is correct because the truncation $|z| \leq M^{1/12}$ is symmetric. The conclusion is moreover confirmed to second order by computation: the C^2 remainder of (4) in the note equals $q(x)/M + O(M^{-2})$ with the classical correction $q = \kappa_4/(8v^2) - 5\kappa_3^2/(24v^3)$, and M times the remainder and its first two derivatives converge (Table 2).

A8. Lemma 1, consequences (5) (correct). With $\log g_M = -MI + \frac{1}{2}\log\{M/(2\pi v)\} + O_{C^2}(M^{-1})$ one gets $\|\phi_M - I\|_{C^2(J)} = O(M^{-1}\log M)$ and $\|\phi_M' - I'\|_{C^1(J)} = O(M^{-1})$.

A9. Lemma 2, the limit function H (correct, one step brief). Continuity, $H(0^+) = 0$, strict increase and $H(a) \to \infty$ as $a \uparrow \frac{1}{2}$ are correct. The divergence deserves one line: $\Lambda'(s) = \frac{1}{2}\coth(s/2) - 1/s$ gives $I'(x) \sim (\frac{1}{2} - x)^{-1}$, and $\frac{1}{2} - \frac{1}{2}\sin\theta \sim \frac{1}{4}(\frac{\pi}{2} - \theta)^2$ makes $\int^{\pi/2} d\theta/(1 - \sin\theta)$ diverge. Strict increase and continuity are formalised (`Hint_strictMonoOn`, `Hint_continuousOn`).

A10. Lemma 2, statement for small M (minor). As literally stated ("for every K and $t = N/M \in K$ there exists a unique $a_{N,M}$") the lemma includes $M = 1$, where $g_1 \equiv 1$ on its support, $\phi_1' \equiv 0$ and no positive root exists. For $M = 2$ a unique root exists, since $x\phi_2'(x) = |x|/(1 - 2|x|)$ is continuous and increasing, but Q' is then discontinuous at 0, so Lemma 3 needs $M \geq 3$. The statement should read "for every $M \geq 2$", and the estimate (9) of the note holds for $M \geq M_0(K)$.

A11. Lemma 2, unimodality and strict log-concavity (correct; reference clarified). The induction $f'_{M+1}(x) = f_M(x+\frac{1}{2}) - f_M(x-\frac{1}{2}) < 0$ for $0 < x < (M+1)/2$ is correct with base case $M = 2$ (not $M = 1$). The argument that $\log f_M$ is not affine on any subinterval is correct, because a polynomial of the form $e^{\lambda x+\mu}$ on an interval is constant, and f_M has exact degree $M - 1 \geq 1$ on every knot interval (the leading coefficient on the $(k+1)$th interval is $(-1)^k\binom{M-1}{k}/(M-1)! \neq 0$). Floater [3, Lemma 2, (11) and the discussion after it] gives more: for degree $d = M - 1 \geq 2$ the B-spline is d-concave, inequality (11) is strict on the interior knot intervals, and on the two end intervals the spline is a pure power. Since $ff'' - (f')^2 = \{ff'' - \frac{d-1}{d}(f')^2\} - \frac{1}{d}(f')^2$, this gives $(\log f_M)'' < 0$ on every open knot interval, which is exactly what the note needs. It makes ϕ'_M continuous and strictly increasing on the open support for $M \geq 3$, where $f_M \in C^1$.

A12. Lemma 2, estimate (9) (substantial for the write-up, now formalised). The note's proof omits the a priori localisation of $a_{N,M}$ in a compact subinterval where H' is bounded below. The complete argument, with explicit constants, is formalised in Lean (`endpoint_lemma`): if $|\phi'_M - I'| \leq C/M$ on $[0, b]$ and $H(\mathrm{lo}) + bC/(2M) < t_1 \leq t_2 < H(\mathrm{hi}) - bC/(2M)$, then for every $t \in [t_1, t_2]$ the finite-M root exists, is unique, lies in $(\mathrm{lo}, \mathrm{hi})$ and satisfies $|a_{N,M} - a_t| \leq 2bC/(m\,\mathrm{lo}\,M)$, where m is a lower bound for I'' on $[0, b]$.

A13. Lemma 3, conditions (i)–(iv) of Definition 1.1 (correct, two corrections). I checked Definition 1.1 [5, pp. 6–7] directly. Condition (i) holds because $Q_{N,M} \to +\infty$ at the finite endpoints $\pm 1/(2a)$. Condition (ii) requires $M \geq 3$. Conditions (iii) and (iv) follow from a uniform bound on ϕ''_M over $[-ar^*, ar^*]$ (Lemma 1) by the chain rule and Lipschitz $\Rightarrow$ Hölder (both formalised: `Qprime_lipschitz`, `holder_of_lipschitz`). Two corrections are needed. First, r^* must satisfy $r^* \sup a_{N,M} < \frac{1}{2}$, not merely $r^* a_+ < \frac{1}{2}$; take $r^*(a_+ + \delta) < \frac{1}{2}$ and M so large that $a_{N,M} \leq a_+ + \delta$ (Lemma 2). Second, Definition 1.1 demands the conditions for *every* $n \geq 1$ with the same constants, so the finitely many small n and the degrees n that do not occur must be filled with admissible weights; see A16.

A14. Lemma 3, condition (v) and the MRS equations (correct). Remark (b) after Definition 1.1 [5, p. 6] states that the support condition is equivalent to (1.13)–(1.14). With $e^{-2NQ} = a g_M(ay)$ one has $Q'(y) = \frac{aM}{2N}\phi'_M(ay)$ (formalised: `Q_hasDerivAt`). I verified, and formalised in the arcsine form (`cos_odd_eq_two_sin`, `mrs_first_iff`), that the first MRS equation is equivalent to the endpoint equation (8) of the note, and that the second holds by parity (`mrs_second`). The normalisation is therefore correct. The change of variable $y = \cos\theta$ in the improper integrals of (1.13)–(1.14) is elementary but not formalised.

A15. Proof of the theorem, application of Theorem 13.4 (correct; a misprint in the source). I read Theorems 13.1–13.4 [5, pp. 123–124] and the proof of Theorem 13.4 [5, p. 138]. The recurrence there is $xp_{n,m} = A_{n,m}p_{n,m+1} + B_{n,m}p_{n,m} + A_{n,m-1}p_{n,m-1}$ with $A_{n,m} = \gamma_{n,m}/\gamma_{n,m+1}$, and (13.8) holds uniformly for $|m-n| \le n^{1/3-\varepsilon}$, the window (13.3). The coefficient linking degrees $N-1$ and N is therefore $A_{N,N-1}$, as in the note, and $m = N-1$ lies in the window. Theorem 2.3(c) [5, p. 12] prints the recurrence with a misprint (the term $A_{n,m}p_{n,m-1}$ should read $A_{n,m}p_{n,m+1}$); the note's reading agrees with Chapter 13. The scaling (10) of the note, $\beta_N^{[M]}/M^2 = a_{N,M}^2 A_{N,N-1}^2$, is verified in Section 3.1 and its algebra is formalised (`beta_scaling`).

A16. Proof of the theorem, uniformity (correct in substance, details missing). The contradiction argument is valid. The "filling" needs two explicit facts: (a) for every large n there is M with n/M in a fixed interval $[t_1, t_2] \subset (0, \infty)$, and (b) the remaining small n can be given a fixed admissible weight, for instance $Q_n(x) = x^2$ on $\mathbb{R}$, for which (1.13) reads $\frac{1}{\pi}\int_{-1}^{1} 2x^2(1-x^2)^{-1/2}dx = 1$. Both the abstract argument and fact (a) are formalised (`uniform_of_sequential`, `fill_ratio`).

A17. Corollary (correct). $\Lambda(s) = s^2/24 - s^4/2880 + O(s^6)$, $I'(x) = 12x + \frac{144}{5}x^3 + O(x^5)$, $H(a) = 3a^2 + \frac{27}{5}a^4 + O(a^6)$ and $F(\alpha) = \frac{\alpha}{12} - \frac{\alpha^2}{20} + O(\alpha^3)$ are all correct; the last two steps are formalised (`Hint_expansion`, `inversion`). Monotonicity, $0 < F < \frac{1}{16}$ and $F(\alpha) \to \frac{1}{16}$ follow from Lemma 2. The note calls these "checks at the two ends of the scaling family". They are properties of F only; the theorem does not imply the fixed-N or fixed-M limits, because it is uniform only on compact ratio sets.

A18. Novelty section (correct, one understatement). The description of the fixed-M theory and of Milovanović's conjectures is accurate. In fact the Taylor coefficients of F agree with the top-degree parts of *all five* coefficients in Conjecture 2, not only the first two (Section 5).

3 Independent proof and necessary repairs.

Throughout, $J = [-b, b]$ with $0 < b < \frac{1}{2}$, $s(x)$ is the tilt with $\Lambda'(s(x)) = x$, $v = \Lambda'' \circ s$, $\kappa_j = \Lambda^{(j)} \circ s$, and constants C, c depend only on b (and later on K).

3.1 Normalisations.

The weight of $y = Y_M/a$ with $a = a_{N,M}$ is $ag_M(ay)$ on $|y| < 1/(2a)$, and $X_M = May$. If P_n^y are the monic orthogonal polynomials of y with $h_n^y = \|P_n^y\|^2$, then $P_n^X(X) = (Ma)^n P_n^y\{X/(Ma)\}$ are those of X_M and $h_n^X = (Ma)^{2n}h_n^y$. Since $\beta_N = h_N/h_{N-1}$ for a monic recurrence, and the orthonormal leading coefficients are $\gamma_n = (h_n^y)^{-1/2}$ with $A_{N,N-1} = \gamma_{N,N-1}/\gamma_{N,N}$ in the convention of [5, p. 124 and p. 138],

$$\beta_N^{[M]} = \frac{h_N^X}{h_{N-1}^X} = (Ma)^2 \frac{h_N^y}{h_{N-1}^y} = (Ma)^2 A_{N,N-1}^2. \tag{3.1}$$

Multiplying the weight by a constant (Levin–Lubinsky's normalisation $Q_n(t_n) = 0$, Remark (f) after Definition 1.1) changes neither β nor A. The external field is

$$Q_{N,M}(y) = \frac{M}{2N}\,\phi_M(ay) - \frac{\log a}{2N}, \qquad Q'_{N,M}(y) = \frac{aM}{2N}\,\phi'_M(ay), \qquad \phi_M = -\frac{1}{M}\log g_M, \tag{3.2}$$

and, with $y = \cos\theta$ and the oddness of ϕ'_M, the first MRS equation becomes

$$\frac{1}{\pi}\int_0^{\pi} \cos\theta\, Q'_{N,M}(\cos\theta)\, d\theta = \frac{M}{N}\cdot\frac{1}{\pi}\int_0^{\pi/2} a\sin\theta\, \phi'_M(a\sin\theta)\, d\theta = 1, \tag{3.3}$$

which is the note's endpoint equation with $t = N/M$. Equations (3.1)–(3.3) are formalised (Section 6). The limit $A_{N,N-1} \to \frac{1}{2}$ is the Chebyshev value for $[-1,1]$, so $\beta_N^{[M]}/M^2 \to a^2/4$; as a sanity check, $\alpha \to 0$ gives $F(\alpha) \sim \alpha/12$, i.e. $\beta_N \approx NM/12 = N\operatorname{Var} X_M$ (Hermite), and $\alpha \to \infty$ gives $\frac{1}{16}$, i.e. $\beta_N \approx M^2/16 = (\text{length}/4)^2$.

3.2 Lemma 1: a complete proof.

Lemma 1. *For every $b < \frac{1}{2}$ there are M_0 and C such that for $M \ge M_0$*

$$\Big\| \log g_M + MI - \frac{1}{2}\log\frac{M}{2\pi v} \Big\|_{C^2([-b,b])} \le \frac{C}{M}. \tag{3.4}$$

Proof. Step 0 (representation). Let $A(z) = \int_{-1/2}^{1/2} e^{zt}dt = 2\sinh(z/2)/z$, so that $\int e^{zy} f_M(y)\, dy = A(z)^M$ and $A = e^{\Lambda}$ on $\mathbb{R}$. For real s the probability density $e^{sy} f_M(y) A(s)^{-M}$ has characteristic function $\{A(s+iu)/A(s)\}^M$. Since $|2\sinh\{(s+iu)/2\}| \le 2\cosh(s/2)$ and $|s+iu| \ge |u|$,

$$|A(s+iu)| \le \frac{2\cosh(s/2)}{|u|}, \tag{3.5}$$

so for $M \ge 2$ the characteristic function is integrable, f_M is continuous, and Fourier inversion gives $e^{sy} f_M(y) A(s)^{-M} = \frac{1}{2\pi}\int_{\mathbb{R}} \{A(s+iu)/A(s)\}^M e^{-iuy} du$. Taking $y = Mx$, $s = s(x)$ and using $A(s)^M e^{-sMx} = e^{M\{\Lambda(s) - sx\}} = e^{-MI(x)}$,

$$g_M(x) = \frac{M}{2\pi}\, e^{-MI(x)} J_M(x), \qquad J_M(x) = \int_{\mathbb{R}} G(x,u)^M\, du, \qquad G(x,u) = e^{-iux}\,\frac{A(s(x)+iu)}{A(s(x))}. \tag{3.6}$$

Step 1 (bounds on G). The map $x \mapsto s(x)$ is real-analytic on $(-\frac{1}{2}, \frac{1}{2})$ (inverse function theorem, $\Lambda'' > 0$), and $S = s(J)$ is compact. (a) $|G| \le 1$, and $|G(x,u)| < 1$ for $u \ne 0$ because $G(x,\cdot)$ is the characteristic function of a law with a density; by continuity and compactness, and by (3.5), for every $\varepsilon > 0$ there is $\rho_\varepsilon < 1$ with $|G| \le \rho_\varepsilon$ for $x \in J$, $|u| \ge \varepsilon$, and $|G| \le C_1/|u|$ for $|u| \ge 1$. (b) Write

$$\partial_x G = -iuG + s'(x)\, e^{-iux}\,\frac{A'(s+iu)A(s) - A(s+iu)A'(s)}{A(s)^2}. \tag{3.7}$$

Since $A'(z) = \cosh(z/2)/z - 2\sinh(z/2)/z^2$, we have $|A'(s+iu)| \le C/|u|$ for $|u| \ge 1$, and $|uG| \le C$; hence $|\partial_x G| \le C$. Differentiating (3.7) once more, the only term not bounded by a constant is $-iu\,\partial_x G$, so $|\partial_x^2 G| \le C(1+|u|)$.

Step 2 (outer region). For $j \le 2$, $\partial_x^j(G^M)$ is a combination of $MG^{M-1}\partial_x^j G$ and $M(M-1)G^{M-2}(\partial_x G)^2$, so $|\partial_x^j(G^M)| \le CM^2(1+|u|)|G|^{M-2}$. For $|u| \ge \varepsilon$ and $M \ge 7$, $|G|^{M-2} \le \rho_\varepsilon^{M-6}|G|^4$ and $(1+|u|)|G|^4$ is integrable by (a). This majorant, uniform in $x \in J$, justifies differentiating J_M twice under the integral and gives

$$\sup_{x\in J}\Big|\partial_x^j \int_{|u|\ge\varepsilon} G^M\, du\Big| \le CM^2\rho_\varepsilon^{M-6}, \qquad j = 0,1,2. \tag{3.8}$$

Step 3 (inner region). Choose $\varepsilon \in (0,1]$ so small that $A(\sigma+iu)$ has positive real part for σ in a neighbourhood S' of S and $|u| \le \varepsilon$ (possible because $A > 0$ on $\mathbb{R}$). Then $\Psi(x,u) =$

$\Lambda(s(x)+iu)-\Lambda(s(x))-iux$, with the principal logarithm, satisfies $G^M = e^{M\Psi}$ there, is real-analytic in x and analytic in u, and

$$\Psi = -\frac{v}{2}u^2 - \frac{i\kappa_3}{6}u^3 + \frac{\kappa_4}{24}u^4 + \Psi_5, \qquad |\partial_x^j \Psi_5| \le C|u|^5, \qquad j \le 2,\ |u| \le \varepsilon, \tag{3.9}$$

because the x-derivatives of a Taylor remainder are the Taylor remainders of the x-derivatives, which are bounded uniformly by Cauchy's estimates. Shrinking ε, $\Re\Psi \le -cu^2$. By the cancellations of A5 (formalised), $|\partial_x\Psi| + |\partial_x^2\Psi| \le Cu^2$. Put $u = z/\sqrt{M}$ and $Z = M^{1/12}$.

On $Z < |z| \le \varepsilon\sqrt{M}$, $|\partial_x^j e^{M\Psi}| \le C\{1 + M(|\partial_x\Psi| + |\partial_x^2\Psi|) + M^2|\partial_x\Psi|^2\}e^{M\Re\Psi} \le C(1+z^4)e^{-cz^2}$, whose integral is $O(e^{-cZ^2/2}) = O(e^{-cM^{1/6}/2})$.

On $|z| \le Z$, write $M\Psi = -\frac{v}{2}z^2 + w$ with

$$w = -\frac{i\kappa_3 z^3}{6\sqrt{M}} + \frac{\kappa_4 z^4}{24M} + M\Psi_5\Big(x, \frac{z}{\sqrt{M}}\Big), \qquad |\partial_x^j w| \le C\,\frac{(1+|z|)^5}{\sqrt{M}} \le CM^{-1/12}, \qquad j \le 2. \tag{3.10}$$

Let $E = e^w - 1 - w - \frac{w^2}{2}$. By Taylor's formula for the exponential and the chain rule, for $j \le 2$, $|\partial_x^j E| \le C\sum |w|^{3-k}\,\Pi_k$, where Π_k is a product of k factors $|\partial_x^i w|$, $i \le 2$; each term is at most $C(1+|z|)^{15}M^{-3/2}$. Expanding $w + \frac{w^2}{2}$ and moving every term of order $M^{-3/2}$ into the remainder,

$$e^{M\Psi} = e^{-vz^2/2}\Big[1 - \frac{i\kappa_3 z^3}{6\sqrt{M}} + \frac{1}{M}\Big(\frac{\kappa_4 z^4}{24} - \frac{\kappa_3^2 z^6}{72}\Big) + \mathcal{E}_M\Big], \qquad |\partial_x^j \mathcal{E}_M| \le C\,\frac{(1+|z|)^{18}}{M^{3/2}}, \tag{3.11}$$

for $j \le 2$ and $|z| \le Z$. The x-derivatives falling on $e^{-vz^2/2}$ produce polynomial factors in z, so after integration over $|z| \le Z$ the remainder contributes $O(M^{-3/2})$ in $C^2(J)$. The z^3 term and its x-derivatives are odd in z and integrate to zero over the symmetric range. Completing the remaining Gaussian moments to $\mathbb{R}$ costs $O(e^{-cZ^2})$. With $\int z^4 e^{-vz^2/2}dz = 3\sqrt{2\pi}v^{-5/2}$ and $\int z^6 e^{-vz^2/2}dz = 15\sqrt{2\pi}v^{-7/2}$ we obtain

$$J_M = \sqrt{\frac{2\pi}{Mv}}\,(1 + R_M), \qquad R_M = \frac{q}{M} + O_{C^2(J)}\big(M^{-3/2}\big), \qquad q = \frac{\kappa_4}{8v^2} - \frac{5\kappa_3^2}{24v^3}, \tag{3.12}$$

where (3.8) and the middle region were used; both are exponentially small relative to $M^{-1/2}$. Since q is smooth on J, $\|R_M\|_{C^2(J)} \le C/M$.

Step 4. By (3.6) and (3.12), $\log g_M + MI - \frac{1}{2}\log\{M/(2\pi v)\} = \log(1+R_M)$, and $\|\log(1+R_M)\|_{C^2(J)} \le C\|R_M\|_{C^2(J)}$ once $\|R_M\|_{C^0} \le \frac{1}{2}$. This proves (3.4), and indeed the sharper statement that the left side equals $q/M + O(M^{-3/2})$ in $C^2(J)$. □

Remark 1. The sharper statement is confirmed numerically (Table 2). At the points $x = 0.05$, 0.15, 0.25, 0.35 and 0.42 one has $q = -0.151544$, -0.163813, -0.184391, -0.167178 and -0.089240, while M times the left side of (3.4) at $M = 800$ equals -0.15157, -0.16386, -0.18449, -0.16724 and -0.08917.

Remark 2. For the theorem with an $o(1)$ error, only two consequences of Lemma 1 are used: $\sup_{[0,b]}|\phi_M' - I'| \to 0$ (with rate $O(M^{-1})$ for (9) of the note) and $\sup_M \sup_{[-b,b]}|\phi_M''| < \infty$. An alternative route avoids differentiating under the integral sign. The exact identities $f_M'(y) = f_{M-1}(y+\frac{1}{2}) - f_{M-1}(y-\frac{1}{2})$ and $f_M''(y) = f_{M-2}(y+1) - 2f_{M-2}(y) + f_{M-2}(y-1)$ express ϕ_M' and ϕ_M'' through ratios of densities at shifted points. The undifferentiated case $j = 0$ of the proof above, applied to f_{M-k} with $k \le 2$, then suffices.

3.3 Lemma 2: corrected statement.

Lemma 2. *Let $H(a) = \frac{1}{\pi}\int_0^{\pi/2} a\sin\theta\, I'(a\sin\theta)\,d\theta$ and let H_M be defined in the same way with ϕ'_M in place of I'.*

(a) H is continuous and strictly increasing on $[0, \frac{1}{2})$, $H(0) = 0$ and $H(a) \to \infty$ as $a \uparrow \frac{1}{2}$; hence a_t is well defined for every $t > 0$.

(b) For every $M \geq 2$ and $t > 0$ the equation $H_M(a) = t$ has a unique root $a_{t,M} \in (0, \frac{1}{2})$; for $M = 1$ it has none.

(c) For every compact $K \subset (0, \infty)$ there are M_0, C with $|a_{t,M} - a_t| \leq C/M$ for all $t \in K$ and $M \geq M_0$.

Proof. Part (a) is in the note, with the divergence argument of A9. For (b), $x\phi'_M(x)$ is continuous on $[0, \frac{1}{2})$, vanishes at 0 and is strictly increasing (A11 for $M \geq 3$; $x\phi'_2(x) = x/(1-2x)$ for $M = 2$). Near $\frac{1}{2}$ the density is a positive multiple of $(\frac{1}{2} - x)^{M-1}$, so $x\phi'_M(x) \geq c/(\frac{1}{2} - x)$ there and $H_M(a) \to \infty$; the root exists and is unique. For (c), choose $0 < \mathrm{lo} < \mathrm{hi} < b < \frac{1}{2}$ with $H(\mathrm{lo}) < \min K$ and $\max K < H(\mathrm{hi})$, and let $m = \min_{[0,b]} I'' > 0$. Lemma 1 gives $|\phi'_M - I'| \leq C/M$ on $[0, b]$, and $I' \geq 0$, $I'(y') - I'(y) \geq m(y' - y)$ on $[0, b]$. The formal theorem `endpoint_lemma` then yields (c) with constant $2bC/(m\,\mathrm{lo})$. □

3.4 Lemma 3: corrected statement.

Lemma 3. *Let $K \subset (0, \infty)$ be compact. There are N_0 and constants $r^* > 1$, $\alpha \in (0, \frac{1}{4}]$, $C > 0$, $\alpha_1 = \frac{3}{4}$, $C_1 > 0$ and an open neighbourhood I_0 of ± 1 such that, for all (N, M) with $N \geq N_0$, $M \geq 3$ and $N/M \in K$, the weight $e^{-2NQ_{N,M}} = a_{N,M} g_M(a_{N,M}\,\cdot)$ on $I_{N,M} = (-1/(2a_{N,M}), 1/(2a_{N,M}))$ satisfies conditions (i)–(v) of Definition 1.1 of [5] with these constants.*

Proof. Let $a_\pm$ be the extreme values of a_t on K and fix $\delta > 0$ and $r^* > 1$ with $r^*(a_+ + \delta) < \frac{1}{2}$; by Lemma 2, $a_-/2 \leq a_{N,M} \leq a_+ + \delta$ for $N \geq N_0$. Then $[-r^*, r^*] \subset I_{N,M}$ (formalised: `interval_contains`). Condition (i) holds by A13 and condition (ii) by A11. By (3.2), $|Q'_{N,M}(x) - Q'_{N,M}(y)| \leq \frac{a^2 M}{2N} L_M |x - y|$ on $[-r^*, r^*]$, where $L_M = \sup\{|\phi''_M(u)| : |u| \leq \frac{1}{2} - \eta\}$ for $\eta = \frac{1}{2} - r^*(a_+ + \delta)$; L_M is bounded uniformly in $M \geq M_0$ by Lemma 1, and $M/N \leq 1/\min K$. Hence $Q'_{N,M}$ is uniformly Lipschitz on $[-r^*, r^*]$, and therefore uniformly Hölder of orders α and α_1 there; take $I_0 = (-1 - \eta', -1 + \eta') \cup (1 - \eta', 1 + \eta')$ with $1 + \eta' < r^*$. These give (iii) and (iv). Condition (v) is (3.3) together with the second MRS equation, which holds by parity; by Remark (b) after Definition 1.1 these are equivalent to (v). □

3.5 Proof of the theorem.

Let $E(N, M) = \beta_N^{[M]}/M^2 - F(N/M)$, and let $K' = [t_1, t_2] \subset (0, \infty)$ with $t_1 < t_2$ and $K \subset K'$. Take any $m : \mathbb{N} \to \mathbb{N}$ with $n/m(n) \in K'$ for all large n. Define $Q_n = Q_{n,m(n)}$ for $n \geq N_0(K')$ and $Q_n(x) = x^2$ on $\mathbb{R}$ otherwise. By Lemma 3, $\{Q_n\} \in \mathcal{Q}$, so Theorem 13.4 of [5] gives $A_{n,n-1} = \frac{1}{2} + o(1)$. By (3.1), $E(n, m(n)) = a_{n,m(n)}^2 A_{n,n-1}^2 - a_t^2/4$ with $t = n/m(n)$. Since $a_{n,m(n)} - a_t \to 0$ by Lemma 2(c), and F is bounded, $E(n, m(n)) \to 0$. This is the hypothesis `hseq` of the formal theorem `uniform_of_sequential`. The filling hypothesis is `fill_ratio`, so that theorem yields: for every $\varepsilon > 0$ there is L with $|E(N, M)| < \varepsilon$ whenever $N \geq L$ and $N/M \in K$. This is (2) of the note. □

3.6 What remains open.

On paper, nothing remains open: with Sections 3.2–3.5 the theorem is proved modulo the published Theorem 13.4 and Remark (b) after Definition 1.1 of Levin–Lubinsky, which are used as black boxes. No rate is claimed. A rate $O(N^{-\tau})$ would require knowing that the implied constants in Theorem 13.4 depend on $\{Q_n\}$ only through the constants of Definition 1.1. The book does not state this explicitly, and I have not verified it through the proofs.

4 Search for counterexamples and numerical consistency checks.

Method. The weight is a polynomial of degree $M-1$ on each knot interval. Gauss–Legendre quadrature with $n_g \geq N_{\max} + M/2$ nodes per knot interval therefore reproduces the first $N_{\max}$ recurrence coefficients exactly (the discretisation idea of Gautschi [4]). Spline values come from the positive B-spline recurrence, and the discretised Stieltjes procedure is run in 64-bit-mantissa extended precision. Against exact rational values (moments and the Chebyshev algorithm) the code agrees to about 17 significant digits; for example both give $\beta_{12}^{[40]} = 33.68595680113984\ldots$. All diagonal coefficients vanish to within 10^{-16}. The profile F was computed with `mpmath` at 30 digits. Scripts and logs are in the folder `numerics/uniform_lanczos` of the certification repository.

$t = N/M$	$F(t)$	$M\,E$, $M = 100$	$M\,E$, $M = 200$	$M\,E$, $M = 400$	E, $M = 400$
1/4	0.0178432	0.010828	0.010800	0.010786	2.70×10^{-5}
1/2	0.0303768	0.017171	0.017111	0.017081	4.27×10^{-5}
1	0.0445361	0.017648	0.017602	0.017579	4.39×10^{-5}
2	0.0548091	0.010479	0.010471	0.010466	2.62×10^{-5}
4	0.0598372	0.004340	0.004341	0.004342	1.09×10^{-5}

Table 1: $E = \beta_N^{[M]}/M^2 - F(N/M)$ with $N = tM$ (largest case $N = 1600$). ME is essentially constant in M, i.e. $E = c(t)/M + O(M^{-2})$.

x	$q(x)$	MR at $M = 800$	MR', $M = 50 \to 800$	MR'', $M = 50 \to 800$
0.05	−0.151544	−0.15157	$-0.06315 \to -0.06186$	$-1.2664 \to -1.2388$
0.25	−0.184391	−0.18449	$-0.19104 \to -0.18107$	$2.2461 \to 2.2481$
0.42	−0.089240	−0.08917	$0.50738 \to 0.56497$	$-35.856 \to -35.183$

Table 2: $R = \log g_M + MI - \frac{1}{2}\log\{M/(2\pi v)\}$ and its derivatives, computed exactly through $f'_M = \Delta f_{M-1}$ and $f''_M = \Delta^2 f_{M-2}$; MR, MR' and MR'' converge, as Lemma 1 predicts, and $MR \to q$.

Counterexample search. I looked for failures of uniformity near particular ratios (t from $\frac{1}{4}$ to 4), for normalisation errors (a wrong factor would show up as a non-vanishing limit of E, whereas $E \to 0$ like $1/M$ in every case), and for finite-parameter behaviour incompatible with the limit. I found none. The edge of J ($x = 0.42$) has larger constants, as expected, but the same $1/M$ behaviour. These computations are evidence, not proof.

Milovanović's Conjecture 2. As a by-product, for fixed $k = 3, 6, 10$ the error of the five-term formula of Conjecture 2 behaves like $c_k\, m^{-4}$ for $m = 50$ to 400 (for $k = 3$, $m^4 \times$ error is 0.0100, 0.0099, 0.0099, 0.0098). This supports the conjecture numerically and proves nothing.

5 Originality and bibliography.

(I) Fixed M, $N \to \infty$. For fixed $M \geq 2$ the weight f_M is positive on $(-\frac{M}{2}, \frac{M}{2})$ and vanishes polynomially at the endpoints, so it lies in the Szegő class, and a fortiori Rakhmanov's theorem (see [2]) applies. Hence $\beta_N^{[M]} \to (M/4)^2 = M^2/16$ as $N \to \infty$. This is classical and says nothing uniform in M, as the note correctly states.

(II) Milovanović's conjectures. I read Section 5 of [7] in the original (pp. 345–347). Conjecture 1 asserts that for $k \geq 3$, $\beta_k^{(m)} = C_k\, q_{s_k-3}(m) q_{s_k}(m) / \{q_{s_k-2}(m) q_{s_k-1}(m)\}$ with integer polynomials of degrees $s_k = \{2k^2 - 1 + (-1)^k\}/8$ and rational C_k (an exact-structure statement at fixed k). Conjecture 2 asserts, for fixed $k \geq 1$ and $m \to \infty$,

$$\beta_k^{[m]} \asymp \frac{k}{12}m - \frac{k(k-1)}{20} + \frac{4k(k-1)(k-2)}{525m} + \frac{2k(k-1)(k-2)(5k-9)}{2625m^2} + \frac{2k(k-1)(k-2)(552k^2 - 1279k + 255)}{1010625m^3} + \cdots . \tag{5.1}$$

The theorem under review proves neither conjecture, and nothing here should be read as claiming that it does. There is, however, an exact formal consistency, which I verified by exact rational arithmetic. The Taylor coefficients of F at 0 are

$$F(\alpha) = \frac{\alpha}{12} - \frac{\alpha^2}{20} + \frac{4\alpha^3}{525} + \frac{2\alpha^4}{525} + \frac{368\alpha^5}{336875} - \frac{544\alpha^6}{1340625} + O(\alpha^7), \tag{5.2}$$

and the first five coincide with the coefficients of the top-degree monomials $k^j m^{2-j}$ in the five terms of (5.1); note that $\frac{2\cdot 5}{2625} = \frac{2}{525}$ and $\frac{2\cdot 552}{1010625} = \frac{368}{336875}$. This is what one expects if the fixed-k expansion and the ratio-scaling limit match in an overlap regime $1 \ll k \ll m$. It is a non-trivial check of both, but it does not prove either, because no uniformity bridging the two regimes is available. If Conjecture 2 continues with further terms, (5.2) predicts the top-degree coefficient $-544/1340625$ of the next one.

(III) The two-parameter theorem. The general mechanism, that recurrence coefficients are asymptotic to half the Mhaskar–Rakhmanov–Saff number, goes back to Freud's conjecture for fixed exponential weights [6]. For varying weights the tool is [5]. The theorem is therefore a new *application* of an established theorem, not a new general theorem. Its specific content is: the identification of the external field with the Cramér rate function I of the uniform law; the uniform C^2 saddlepoint control of the cardinal B-spline family (Lemma 1, of the classical type initiated by Daniels [1]); and the explicit profile F. I searched the web, Springer and arXiv, starting from [7] and Gautschi's companion paper [4]. I found no earlier statement of this joint limit or of F. The abstract of [4] concerns computation only; its full text was not accessible to me. A limited search cannot establish priority, so the result should be described as "to the author's knowledge new".

6 Formal-machine certification (Lean 4).

Environment. Lean `v4.35.0`, Mathlib commit `0272bbe1c093d038d7257ab15306abd58245f9ff` (the same as for the certification repository of V478). Library `UniformLanczos` (six modules) in the repository `ralfanop/Certifications_Eulerian_polar-curve_computations`, branch `hjm2026-check`, folder `lean/UniformLanczos`. Every file compiles with no error and no `sorry`, and no `native_decide` is used. All 30 main declarations depend only on Lean's standard axioms `propext`, `Classical.choice` and `Quot.sound` (file `lean/ULAxiomCheck.lean`, log `lean/logs/uniform_lanczos_axioms.log`). A clean build of the library takes about 40 seconds.

What is machine-checked. The following list gives the module and the main declarations for each part of the note.

1. `Cancellation`: the derivative cancellations of Lemma 1 (A5) and Legendre duality $I' = s$, $I'' = 1/v$, for arbitrary Λ with the stated derivatives (`tilt_first`, `tilt_second`, `tilt_cancellations`, `legendre_deriv`, `legendre_second`).

2. `Endpoint`: all of Lemma 2(c) and the monotonicity and continuity in Lemma 2(a)–(b), in the θ-form of H (`Hint_strictMonoOn`, `Hint_continuousOn`, `Hint_slope`, `Hint_close`, `exists_unique_root`, `root_stability`, `endpoint_lemma`).

3. `ClassQ`: the elementary parts of Lemma 3, namely the formula for Q', strict monotonicity, Lipschitz transfer, Lipschitz $\Rightarrow$ Hölder and $[-r^*, r^*] \subset I_{N,M}$ (`Q_hasDerivAt`, `Qprime_strictMono`, `Qprime_lipschitz`, `holder_of_lipschitz`, `interval_contains`).

4. `MRS`: the equivalence of the first MRS equation (arcsine form) with the endpoint equation, the second MRS equation by parity, and the scaling $\beta = (Ma)^2 A^2$ (`cos_odd_eq_two_sin`, `mrs_first_iff`, `mrs_second`, `beta_scaling`).

5. `Diagonal`: the uniformity argument of the theorem and the filling lemma (`uniform_of_sequential`, `fill_ratio`).

6. `Corollary`: $\int_0^{\pi/2} \sin^4 = 3\pi/16$, the expansion $H(a) = 3a^2 + \frac{27}{5}a^4 + O(a^6)$ from that of I', and the inversion to $F(\alpha) = \frac{\alpha}{12} - \frac{\alpha^2}{20} + O(\alpha^3)$ (`Hint_expansion`, `A_le_half_alpha`, `inversion`).

What is *not* machine-checked. The following inputs are hypotheses of the formal statements, not formal theorems.

1. The analytic content of Lemma 1, i.e. the representation (3.6), the bounds on G and the Laplace-method estimates. Consequently the hypotheses $|\phi'_M - I'| \le C/M$ and $\sup |\phi''_M| < \infty$ are not formal.

2. The facts about the concrete Λ: $\Lambda'' > 0$, $\Lambda'(\pm\infty) = \pm\frac{1}{2}$, $H(a) \to \infty$, and the expansion of I' at 0.

3. The log-concavity of B-splines (Floater).

4. The substitution $y = \cos\theta$ in the improper MRS integrals.

5. All potential theory, i.e. Remark (b) after Definition 1.1 and Theorem 13.4 of Levin–Lubinsky. Neither is in Mathlib, and formalising them is far beyond the scope of this review.

Hence the three lemmas are *not* formalised in full. Lemma 2 is formalised completely from its stated inputs. Lemmas 1 and 3 are formalised only in their algebraic and elementary parts. The deduction of the theorem from them is formalised as logic, with Levin–Lubinsky's theorem entering as an explicit hypothesis.

7 Final recommendation.

The theorem is, in my judgement, correct, and with the repairs of Section 3 it has a complete proof on paper conditional on [5, Theorem 13.4]. The note is not ready for incorporation as it stands. Before incorporation, five obligations remain:

1. replace the proof of Lemma 1 by a complete one, such as Section 3.2;
2. restrict Lemma 2 to $M \geq 2$, and Lemma 3 to $M \geq 3$ and $N \geq N_0(K)$;
3. correct the choice of r^* (A13);
4. make the filling of the diagonal sequence explicit (A16);
5. describe the result as a new application of Levin–Lubinsky's theorem, with [7] and [4] cited and the consistency (5.2) with Conjecture 2 stated as consistency only.

Separately, the author's own criterion was that the three lemmas enter the manuscript only if they are formalised and certified. That criterion is *not* met. Lemma 2 is fully machine-checked relative to its analytic inputs. Lemmas 1 and 3 are only partly machine-checked, and their remaining parts depend on Fourier analysis and potential theory that are not available in Mathlib. Whether to include the theorem with a paper proof and an explicit statement of its formal coverage, or to wait for a fuller formalisation, is the author's decision.

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